Validation with Flowering data

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2026-04-20

First using the global data

```r
rm(list=ls())
library(spaMM)
```

```
## spaMM (Rousset & Ferdy, 2014, version 4.6.1) is loaded.
## Type 'help(spaMM)' for a short introduction,
## 'news(package='spaMM')' for news,
## and 'citation('spaMM')' for proper citation.
## Further infos, slides, etc. at https://gitlab.mbb.univ-montp2.fr/francois/spamm-ref.
```

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.0     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.2     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(popbio)

MatrixDim <- 50

spaMM.options(separation_max=70)
load("../../IPM/IPM_Kernal_Year.RData")
load("../../IPM/VitalRates.RData")

annees <- 1995:2021
populations <- c("E2","E1","Au","Po","Pe","Cr")

#Observed number of flowering plants
Nf <- c(494,420,604,638,343,490,631,482,815,435,613,794,210,NA,322,216,277,386,565,254,369,407,148,162,3

#Flowering Probability
#Flowering function for The survival-growth kernel
flr0 <- function(x, a, Beta=0,Pop,year) {
  fake_data <- expand.grid(
  year = year,
  Pop = populations,
```

```
  SizeOMars = unique(x),
  Age = a)
  predi <- plogis(predict(Flowering, newdata = fake_data, allow.new.levels = T, type = "link") +  Beta)
  meanpredi <- aggregate(predi, list(fake_data$SizeOMars), mean)
  return(meanpredi$V1)
}

minsize <-  0.5
maxsize <-  25
maxsize1 <- 15
n.size <- MatrixDim
h <-  (maxsize - minsize) / n.size
h1 <-  (maxsize1 - minsize) / n.size
b = minsize + c(0:n.size) * h

#sizes for an individual of age a
#midpoint
ymid.a = 0.5 * (b[1:n.size] + b[2:(n.size + 1)])
#intervals
yI.a <- matrix(nrow = n.size, ncol = 2)
for (i in 1:n.size){
  yI.a[i,1] <- b[i]
  yI.a[i,2] <- b[i+1]
}

b1 = minsize + c(0:n.size) * h1
#sizes for an individual of age 1
#midpoint
ymid.1 = 0.5 * (b1[1:n.size] + b1[2:(n.size + 1)])
#intervals
yI.1 <- matrix(nrow = n.size, ncol = 2)
for (i in 1:n.size){
  yI.1[i,1] <- b1[i]
  yI.1[i,2] <- b1[i+1]
}
```

Number of observed flowering plants since 1995 with IPM We predict the number of flowering plants by year starting by the observed number of flowering plants the first year.

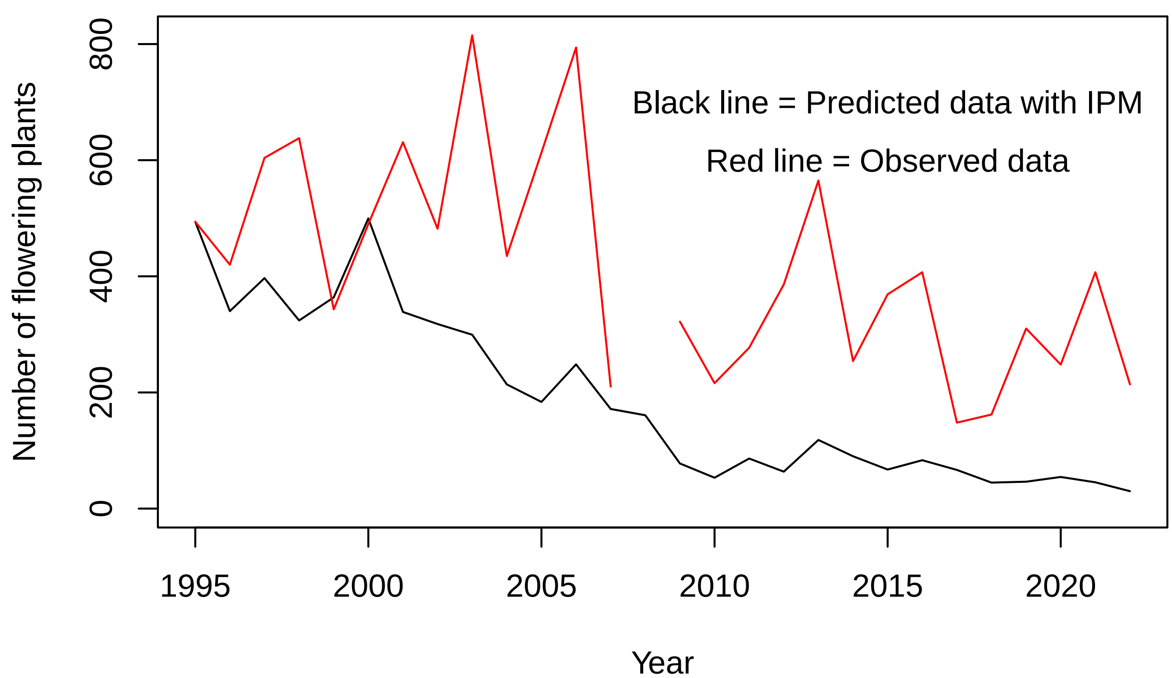
```
N <- vector("double",length(annees)+1)
Nia <- array(data = 0, dim = c(length(annees)+1, 8, MatrixDim))
Nf_est_IPM <- vector("double",length(annees)+1)

#Initial distribution using mean kernal
  A <- eigen.analysis(mean(IPMKernal_Year))
  W <- function(age) {return(A$stable.stage[(50 * (age - 1) + 1):(50 * age)])}
  Nf_dens <- sum(W(1) * flr0(ymid.1, 1, year=1995))
  for (a in 2:8) {
    Nf_dens <- Nf_dens + sum(W(a) * flr0(ymid.a, a, year=1995))
  }

# Number of individuals the 1st year using the number of flowering plants
  N[1] <- Nf[1] / Nf_dens
  # Number of plants in each size class and age at time=1
```

```r
  for (a in 1:8) {
    for (i in 1:MatrixDim) {
      Nia[1, a, ] <- N[1] * W(a)
    }
  }

  Nf_est_IPM[1] <- Nf[1]

  # Iteration over years
  for (yr in 2:length(annees)) {
    year <- annees[yr-1]
    A <- eigen.analysis(IPMKernal_Year[[yr-1]])
    Fec <- function(a) {
      #a = age of the mother plant
      return(IPMKernal_Year[[yr-1]][1:50, ((a - 1) * 50 + 1):(a * 50)])
    }
    P <- function(a) {
      #a = age of the plant at time t
      if (a == 8) {
        return(IPMKernal_Year[[yr-1]][((50*(a-1))+1):(50*a), (50*(a-1)+1):(50*a)])
        }
      return(IPMKernal_Year[[yr-1]][((50*a)+1):(50*(a+1)), (50*(a-1)+1):(50*a)])}

    # Number of plants in each size class and age at time=yr
    for (a in 1:8) { # seedlings
      Nia[yr, 1, ] <- Nia[yr, 1, ] + Fec(a) %*% Nia[yr-1, a, ]
    }
    for (a in 2:8) { # Growth and survival
      Nia[yr, a, ] <- P(a) %*% Nia[yr-1, a - 1, ]
    }
    Nia[yr, 8, ] <- Nia[yr, 8, ] + P(8) %*% Nia[yr-1, 8, ]

    #Nb of flowering plants
    Nf_est_IPM[yr] <- sum(Nia[yr, 1, ] * flr0(ymid.1, 1, year=year))
    for (a in 2:8) {
      Nf_est_IPM[yr] <- Nf_est_IPM[yr] + sum(Nia[yr, a, ] * flr0(ymid.a, a, year=year))
    }
  }

#Last year
    year <- 2021
    yr <- 28
    A <- eigen.analysis(IPMKernal_Year[[yr-1]])
    Fec <- function(a) {
       return(IPMKernal_Year[[yr-1]][1:50, ((a - 1) * 50 + 1):(a * 50)])
    }
    P <- function(a) {
      if (a == 8) {
        return(IPMKernal_Year[[yr-1]][((50*(a-1))+1):(50*a), (50*(a-1)+1):(50*a)])
        }
      return(IPMKernal_Year[[yr-1]][((50*a)+1):(50*(a+1)), (50*(a-1)+1):(50*a)])}

     for (a in 1:8) {Nia[yr, 1, ] <- Nia[yr, 1, ] + Fec(a) %*% Nia[yr-1, a, ]}
```

```r
    for (a in 2:8) {Nia[yr, a, ] <- P(a) %*% Nia[yr-1, a - 1, ]}
    Nia[yr, 8, ] <- Nia[yr, 8, ] + P(8) %*% Nia[yr-1, 8, ]

    #Calcul du nombre de plantes en fleurs
    Nf_est_IPM[yr] <- sum(Nia[yr, 1, ] * flr0(ymid.1, 1, year=year))
    for (a in 2:8) {
      Nf_est_IPM[yr] <- Nf_est_IPM[yr] + sum(Nia[yr, a, ] * flr0(ymid.a, a, year=year))
    }
```

Estimated number of flowering plants by years

```r
plot(y=Nf_est_IPM, x=c(annees,2022), type="l", ylim=c(0,max(c(Nf_est_IPM,Nf),na.rm=TRUE)), xlab="Year",
lines(y=Nf, x=c(annees,2022), col="red")
text(2015,600, "Red line = Observed data")
text(2015,700, "Black line = Predicted data with IPM")
```

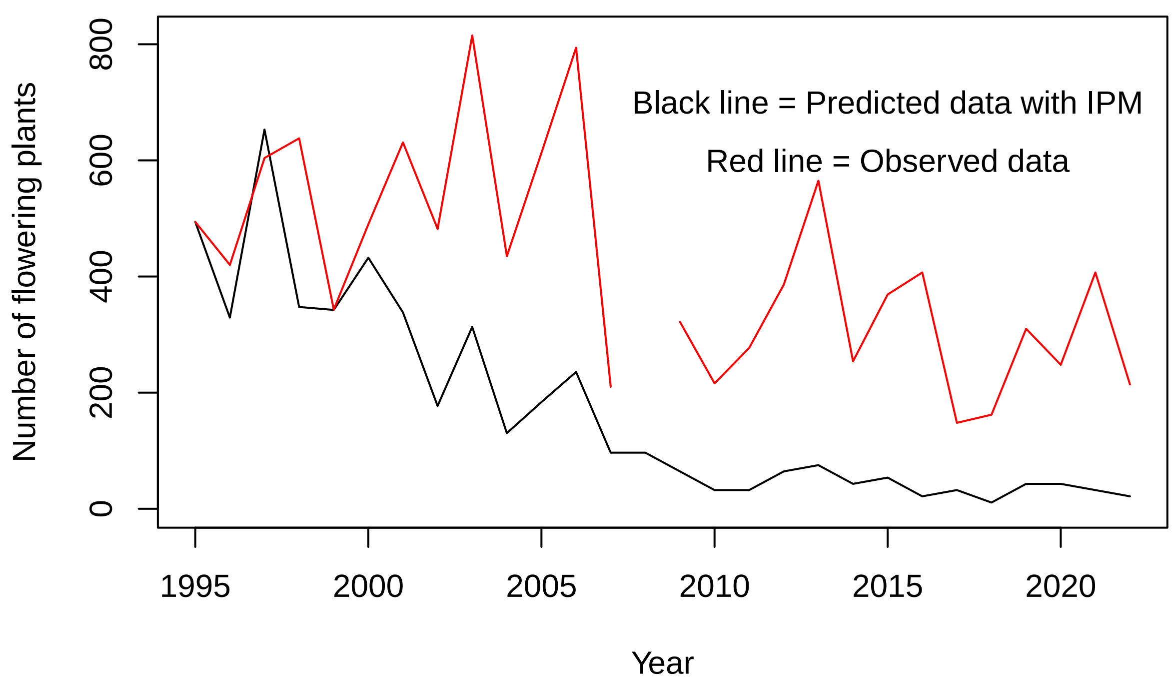


With MPM

```r
load("../MPM/matricesperyear.RData")

Nf_est_MPM <- NA

# Définition de la matrice initiale et du vecteur propre
M <- mean(matrices)  # on prend avec la matrice moyenne pour définir la structure t=0
ev <- eigen(M)
W <- abs(Re(ev$vectors[,1]))
W <- W/sum(W)
N <- round(W*Nf[1]/W[3])
```

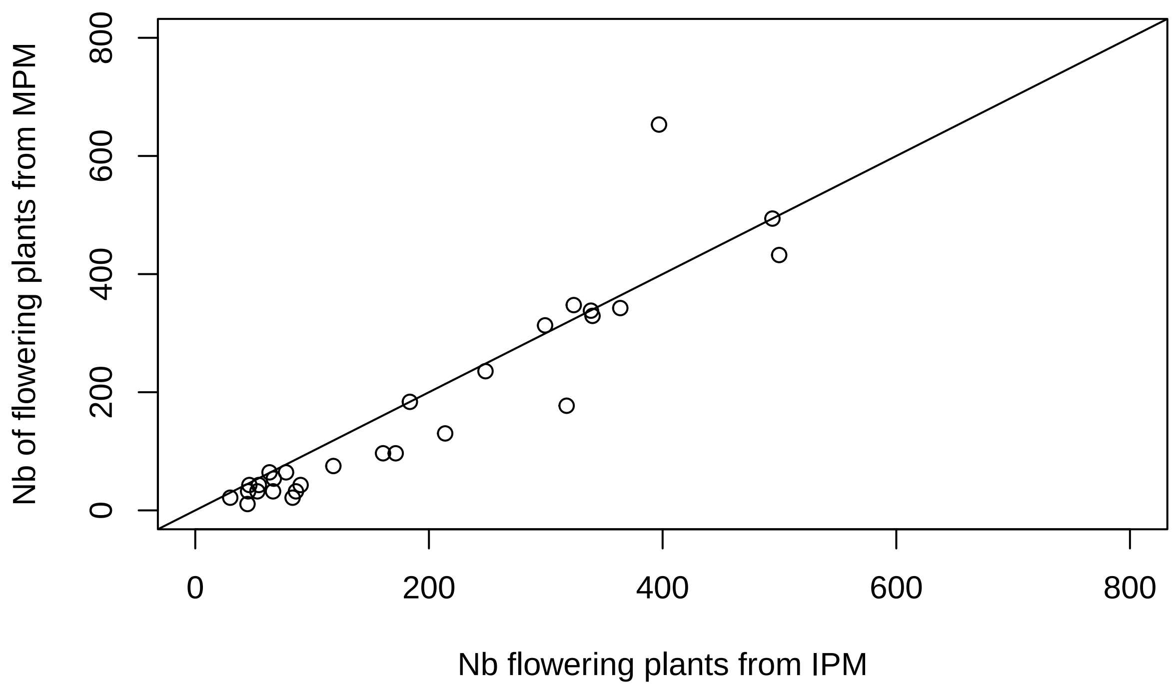
```r
Nf_est_MPM[1] <- Nf[1]

for (i in 2:28){
  M <- matrices[[i-1]]
  ev <- eigen(M)
  N <- M%*%N
  Nf_est_MPM[i] <- N[3]
  }
```

```r
plot(y=Nf_est_MPM, x=c(annees,2022), type="l", ylim=c(0,max(c(Nf_est_MPM, Nf),na.rm=TRUE)), xlab="Year"
lines(y=Nf, x=c(annees,2022), col="red")
text(2015,600, "Red line = Observed data")
text(2015,700, "Black line = Predicted data with IPM")
```



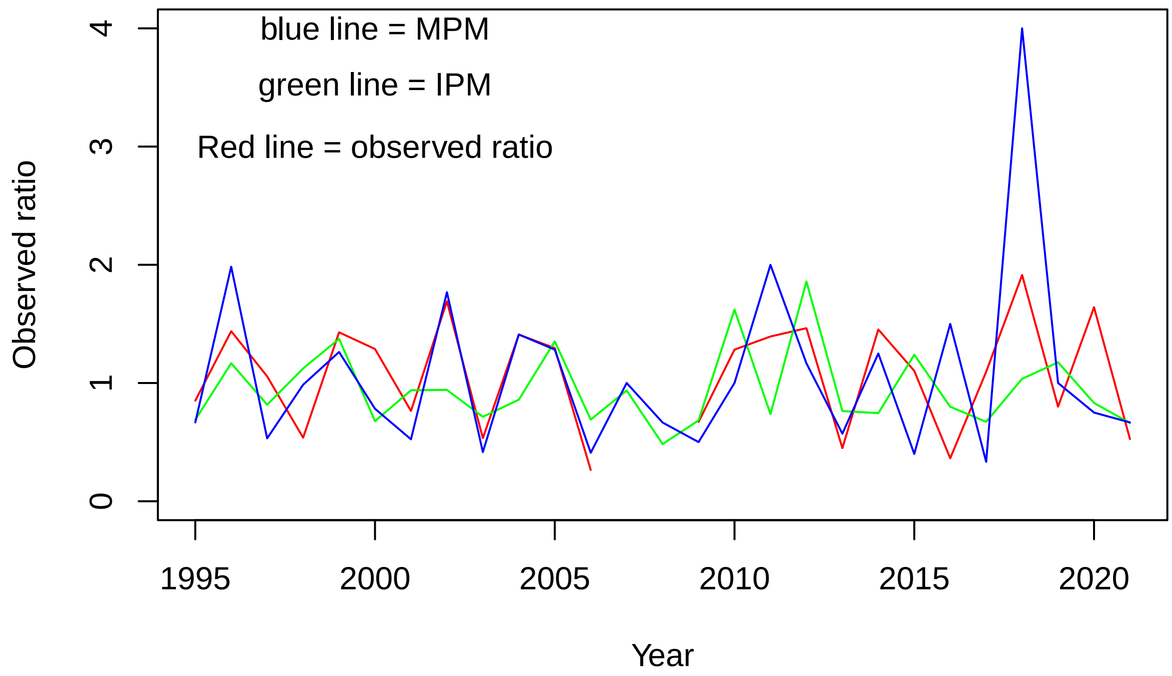
Nf_est_IPM versus Nf_est_MPM

```r
plot(Nf_est_IPM, Nf_est_MPM, xlim=c(0,800), ylim=c(0,800),
     xlab="Nb flowering plants from IPM", ylab="Nb of flowering plants from MPM")
abline(a=0,b=1)
```

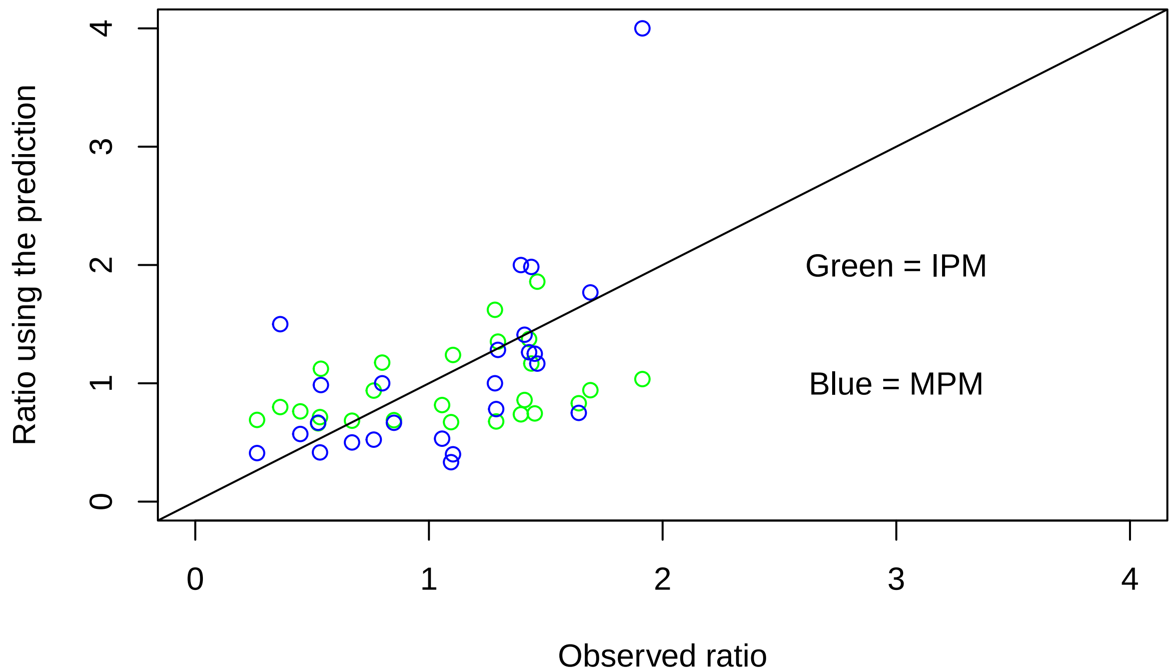


Comparaison des variations interannuelles

```
observed_ratio <- Nf[-1]/Nf[-28]
ratio_IPM <- Nf_est_IPM[-1]/Nf_est_IPM[-28]
ratio_MPM <- Nf_est_MPM[-1]/Nf_est_MPM[-28]

ratio_max <- max(observed_ratio, ratio_IPM,ratio_MPM, na.rm=TRUE)

plot(observed_ratio ~ annees, ylim=c(0,ratio_max), xlab="Year", ylab="Observed ratio", type="l", col="r
lines(ratio_IPM ~ annees, col="green")
lines(ratio_MPM ~ annees, col="blue")
text(2000,3, "Red line = observed ratio")
text(2000,3.5, "green line = IPM")
text(2000,4, "blue line = MPM")
```




```r
plot(observed_ratio, ratio_IPM, xlim=c(0,ratio_max), ylim=c(0,ratio_max),
     xlab="Observed ratio", ylab="Ratio using the prediction", col="green")
points(observed_ratio, ratio_MPM, col="blue")
abline(a=0,b=1)
text(3,1,"Blue = MPM")
text(3,2,"Green = IPM")
```



Stat

```
summary(lm(ratio_IPM ~ observed_ratio))$r.squared
```

```
## [1] 0.1351801
```

```
summary(lm(ratio_MPM ~ observed_ratio))$r.squared
```

```
## [1] 0.3639534
```

```
biaisIPM <- mean(ratio_IPM - observed_ratio, na.rm = TRUE)
print(biaisIPM)
```

```
## [1] -0.1016982
```

```
100 * biaisIPM / mean(observed_ratio, na.rm = TRUE)
```

```
## [1] -9.518576
```

```
biaisMPM <- mean(ratio_MPM - observed_ratio, na.rm = TRUE)
print(biaisMPM)
```

```
## [1] 0.01803372
```

```
100 * biaisMPM / mean(observed_ratio, na.rm = TRUE)
```

```
## [1] 1.687891
```

Same thing but considering each population As many NA values in matrices from MPM in 1995 (age of most plants are unkown), year 1995 is not considered

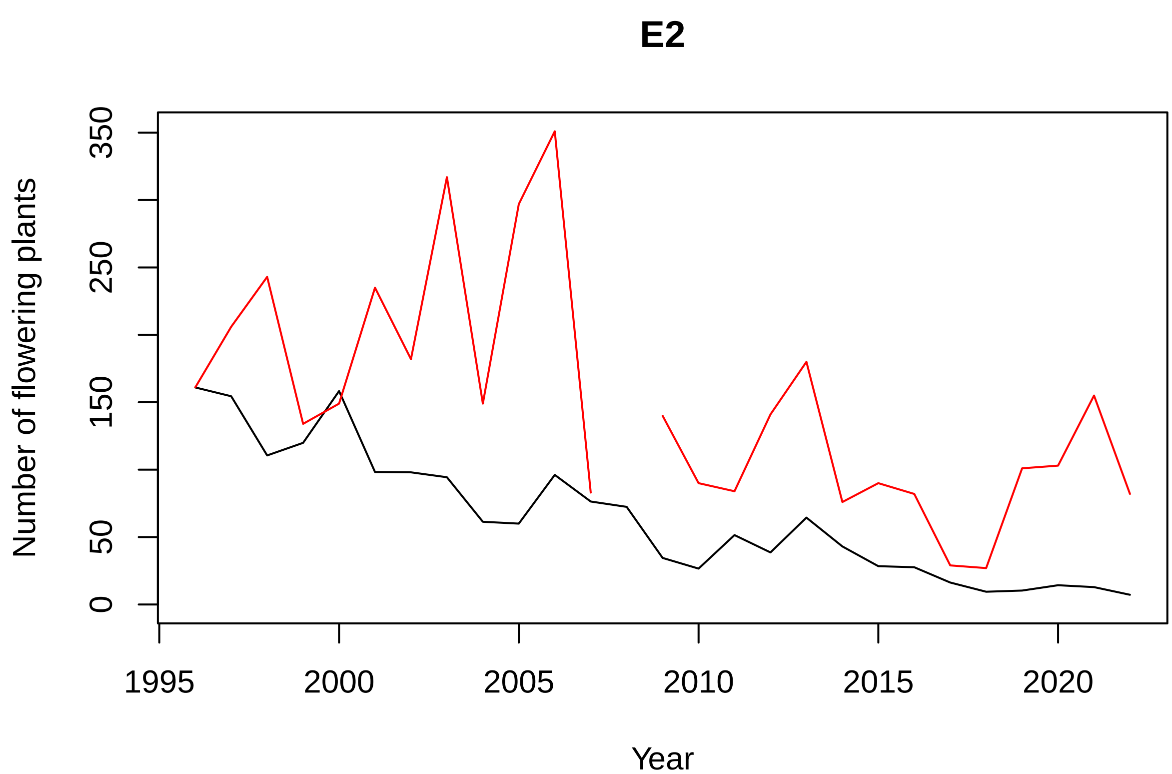
```r
load("../../IPM/IPM_Kernal_Pop_Year.RData")
ME2 <- list();ME1 <- list();MAu <- list();MPo <- list();MPe <- list();MCr <- list()

for (i in 1:27){
MPo[[i]] <- IPMKernalPopYear[[i]];
MAu[[i]] <- IPMKernalPopYear[[i+27]]
MPe[[i]] <- IPMKernalPopYear[[i+27*2]]
ME1[[i]] <- IPMKernalPopYear[[i+27*3]]
ME2[[i]] <- IPMKernalPopYear[[i+27*4]]
MCr[[i]] <- IPMKernalPopYear[[i+27*5]]
}
MPo <- MPo[-1] #remove year=1995
MAu <- MAu[-1]
MPe <- MPe[-1]
ME1 <- ME1[-1]
ME2 <- ME2[-1]
MCr <- MCr[-1]
annees <- 1996:2022
FloweringPlants <- read.csv("flower_data.csv")
FloweringPlants <- FloweringPlants[FloweringPlants$Year>(annees[1]-1) & FloweringPlants$Year<(max(annees


Nf_est_IPM <- matrix(nrow = length(populations),ncol=length(annees))

for (p in 1:length(populations)) {
  # Initialisation
  year <- annees[1]
  Pop <- populations[p]
  Nia <- array(data = 0, dim = c(length(annees), 8, MatrixDim))

  if (Pop=="E2") {matrices <- ME2;Nf <- FloweringPlants$E2}
  if (Pop=="E1") {matrices <- ME1;Nf <- FloweringPlants$E1}
  if (Pop=="Au") {matrices <- MAu;Nf <- FloweringPlants$Au}
  if (Pop=="Po") {matrices <- MPo;Nf <- FloweringPlants$Po}
  if (Pop=="Pe") {matrices <- MPe;Nf <- FloweringPlants$Pe}
  if (Pop=="Cr") {matrices <- MCr;Nf <- FloweringPlants$Cr}
  print(Pop)
  # Initial distribution using the Global_IPM
  A <- eigen.analysis(mean(matrices))
  W <- function(age) {return(A$stable.stage[(50 * (age - 1) + 1):(50 * age)])}

  # Density of flowering plants
  Nf_dens <- sum(W(1) * flr0(ymid.1, 1, Pop=Pop, year=year))
  for (a in 2:8) {Nf_dens <- Nf_dens + sum(W(a) * flr0(ymid.a, a, Pop=Pop, year=year))}

  # Total number of plants 1st year using the observed number of flowering plants
  N <- Nf[1] / Nf_dens
  # Nb of individuals for each size class and age
  for (a in 1:8) {for (i in 1:MatrixDim) {Nia[1, a, ] <- N * W(a)}}

  Nf_est_IPM[p, 1] <- Nf[1]

  # Iteration
```

```
for (yr in 2:length(annees)) {
  year <- annees[yr]
  M <- matrices[[yr-1]]
  A <- eigen.analysis(M)
  W <- function(age) {return(A$stable.stage[(50 * (age - 1) + 1):(50 * age)])}
  Fec <- function(a) {
    #a age de plante mère
    return(M[1:50, ((a - 1) * 50 + 1):(a * 50)])
  }
  P <- function(a) {
    #a age de plante à t
    if (a == 8) {
      return(M[((50*(a-1))+1):(50*a), (50*(a-1)+1):(50*a)])
      }
    return(M[((50*a)+1):(50*(a+1)), (50*(a-1)+1):(50*a)])}

  # Nb of plants in each size class and age
  for (a in 1:8) { # Seedling produced per plants of age a
    Nia[yr, 1, ] <- Nia[yr, 1, ] + Fec(a) %*% Nia[yr - 1, a, ]
  }
  for (a in 2:8) { # Growth-Survival
    Nia[yr, a, ] <- P(a) %*% Nia[yr - 1, a - 1, ]
  }
  Nia[yr, 8, ] <- Nia[yr, 8, ] + P(8) %*% Nia[yr - 1, 8, ]

  #Calcul du nombre de plantes en fleurs
  Nf_est_IPM[p, yr] <- sum(Nia[yr, 1, ] * flr0(ymid.1, 1, Pop=Pop, year=year))
  for (a in 2:8) {
    Nf_est_IPM[p, yr] <- Nf_est_IPM[p, yr] + sum(Nia[yr, a, ] * flr0(ymid.a, a, Pop=Pop, year=year))

 }
}

#last year
  year <- 2022
  M <- matrices[[yr-1]]
  A <- eigen.analysis(M)
  W <- function(age) {return(A$stable.stage[(50 * (age - 1) + 1):(50 * age)])}
  Fec <- function(a) {
    #a age de plante mère
    return(M[1:50, ((a - 1) * 50 + 1):(a * 50)])
  }
  P <- function(a) {
    #a age de plante à t
    if (a == 8) {
      return(M[((50*(a-1))+1):(50*a), (50*(a-1)+1):(50*a)])
      }
    return(M[((50*a)+1):(50*(a+1)), (50*(a-1)+1):(50*a)])}

  # Nb of plants in each size class and age
  for (a in 1:8) { # Seedling produced per plants of age a
    Nia[yr, 1, ] <- Nia[yr, 1, ] + Fec(a) %*% Nia[yr - 1, a, ]
  }
```

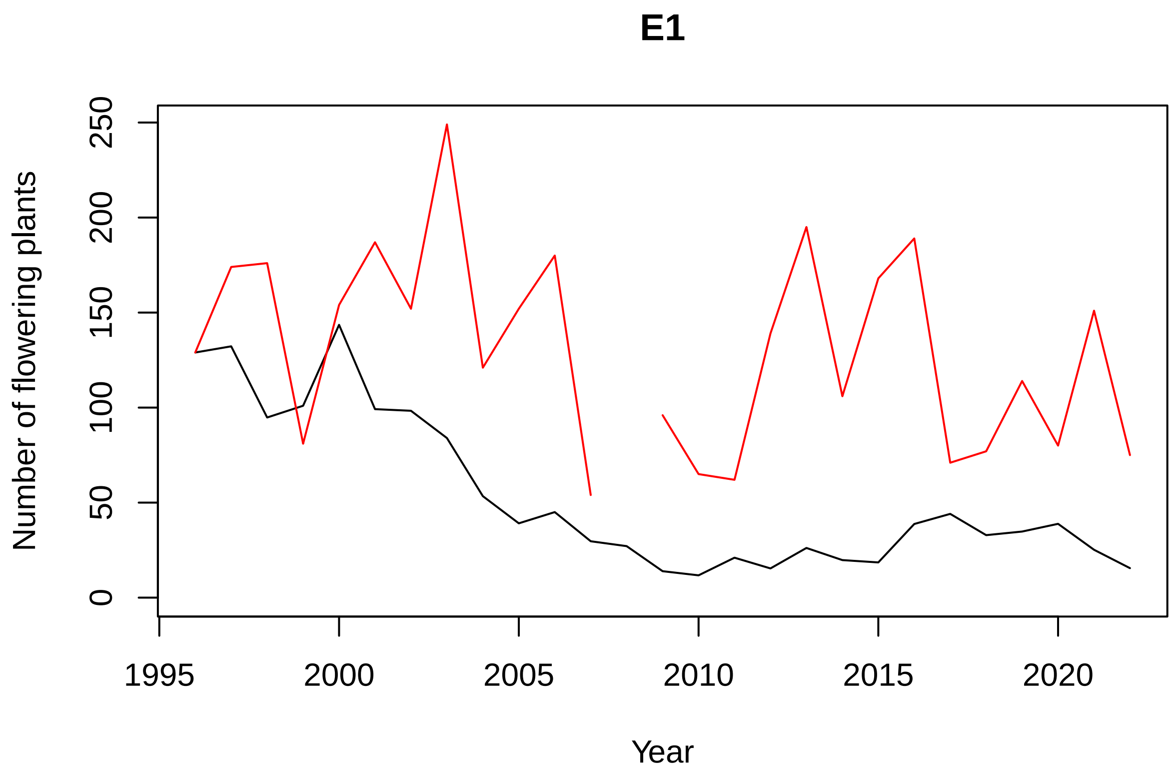
```r
    for (a in 2:8) { # Growth-Survival
      Nia[yr, a, ] <- P(a) %*% Nia[yr - 1, a - 1, ]
    }
    Nia[yr, 8, ] <- Nia[yr, 8, ] + P(8) %*% Nia[yr - 1, 8, ]

    #Calcul du nombre de plantes en fleurs
    Nf_est_IPM[p, yr] <- sum(Nia[yr, 1, ] * flr0(ymid.1, 1, Pop=Pop, year=year))
    for (a in 2:8) {
      Nf_est_IPM[p, yr] <- Nf_est_IPM[p, yr] + sum(Nia[yr, a, ] * flr0(ymid.a, a, Pop=Pop, year=year))


}
}
```

```
## [1] "E2"
## [1] "E1"
## [1] "Au"
## [1] "Po"
## [1] "Pe"
## [1] "Cr"
```

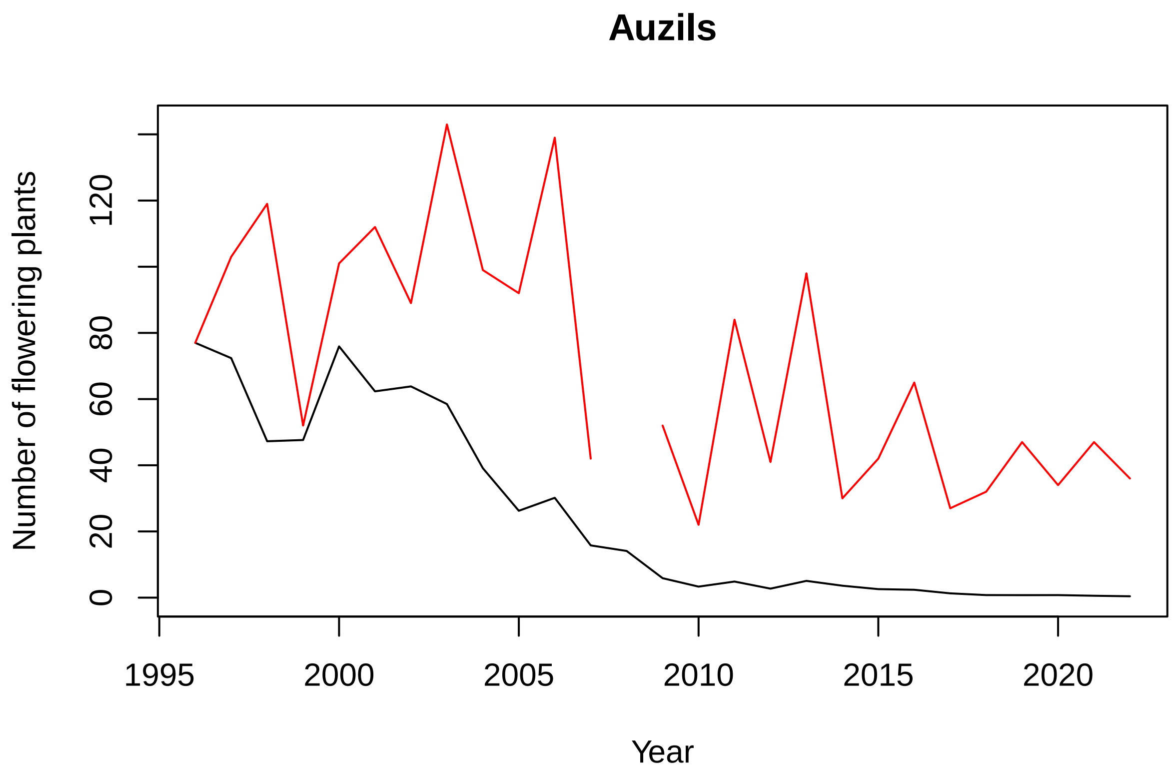
```r
plot(y=Nf_est_IPM[1,], x=annees, type="l", ylim=c(0,max(c(Nf_est_IPM[1,],FloweringPlants$E2),na.rm=TRUE
```



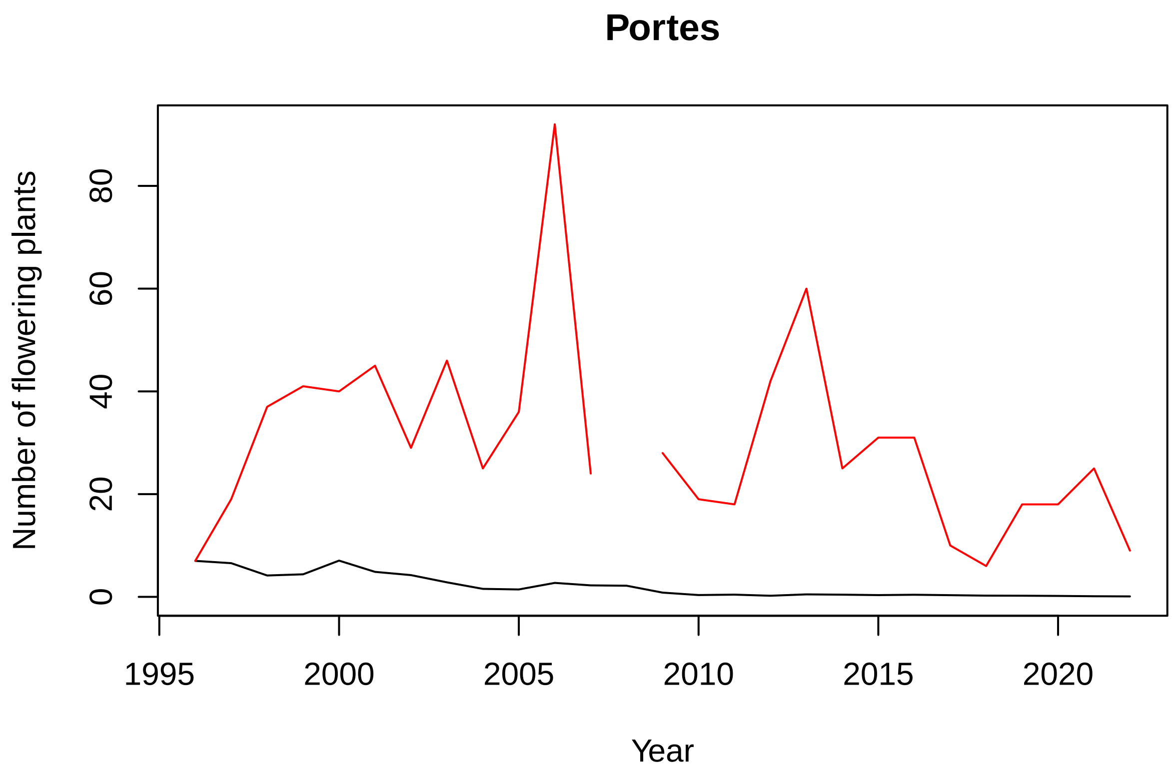

```r
plot(y=Nf_est_IPM[2,], x=annees,type="l",ylim=c(0,max(c(Nf_est_IPM[2,],FloweringPlants$E1),na.rm=TRUE))
```



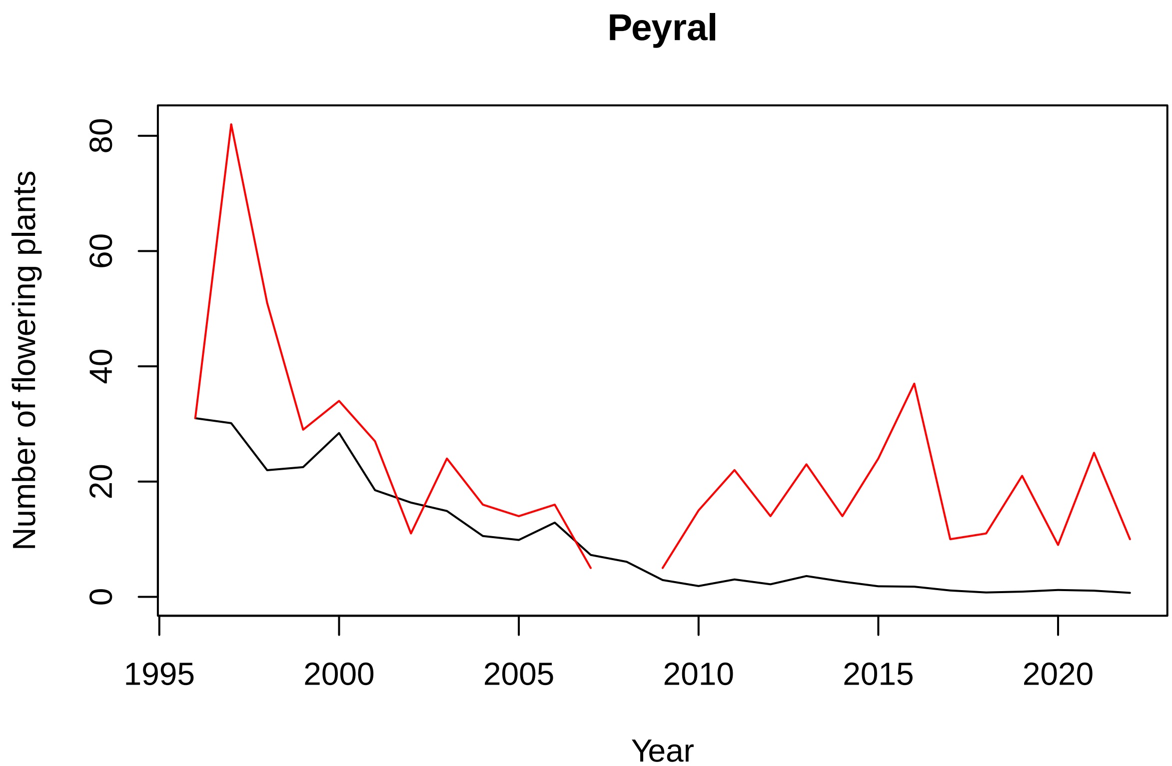

```r
plot(y=Nf_est_IPM[3,], x=annees,type="l",ylim=c(0,max(c(Nf_est_IPM[3,],FloweringPlants$Au),na.rm=TRUE))
```



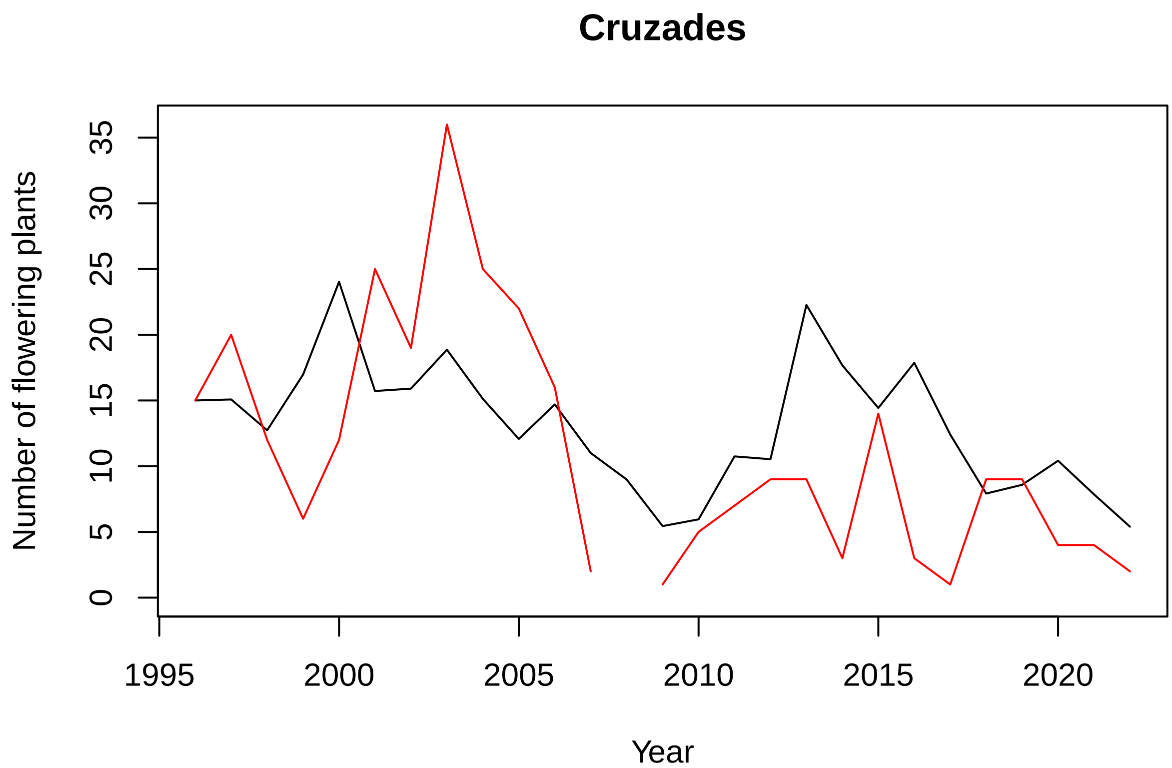

```
plot(y=Nf_est_IPM[4,], x=annees,type="l",ylim=c(0,max(c(Nf_est_IPM[4,],FloweringPlants$Po),na.rm=TRUE))
```




```
plot(y=Nf_est_IPM[5,], x=annees,type="l",ylim=c(0,max(c(Nf_est_IPM[5,],FloweringPlants$Pe),na.rm=TRUE))
```




```
plot(y=Nf_est_IPM[6,], x=annees,type="l",ylim=c(0,max(c(Nf_est_IPM[6,],FloweringPlants$Cr),na.rm=TRUE))
```



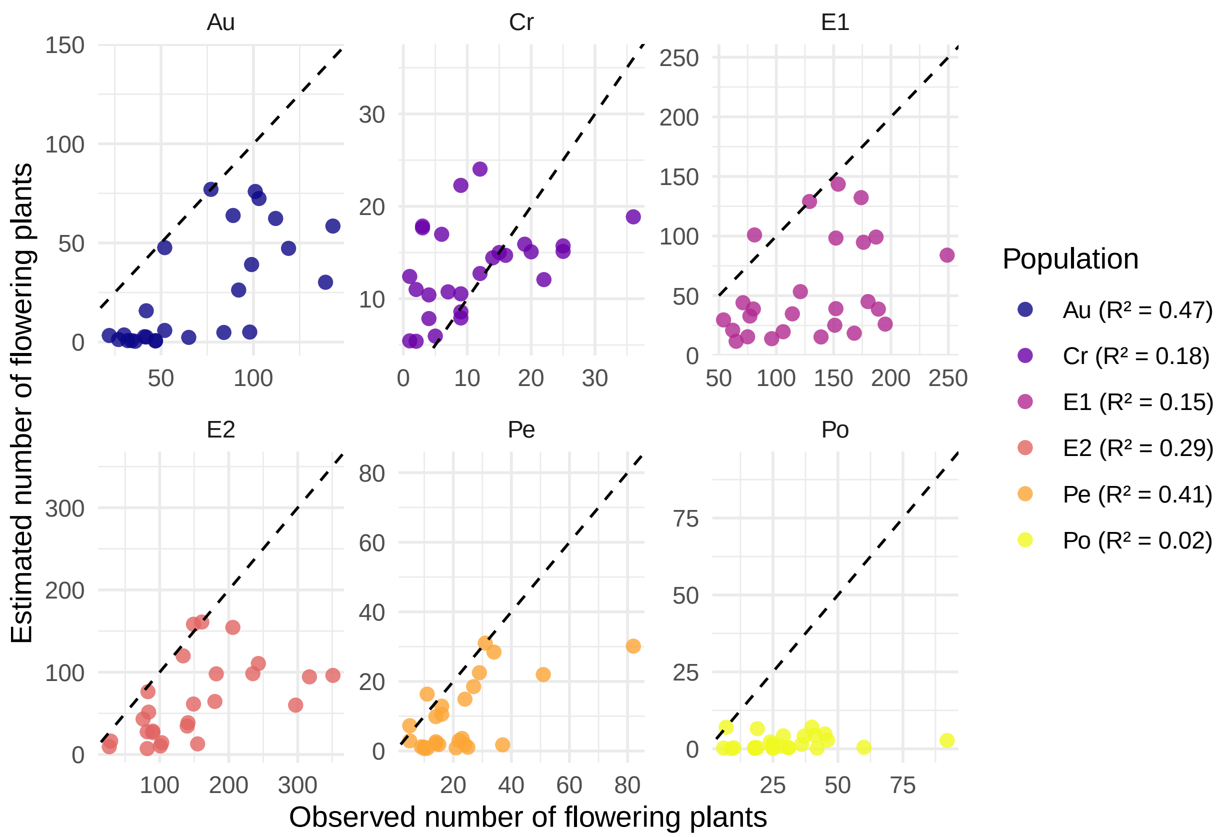

```
Nb_flower_IPM <- expand_grid(
  Pop = populations,
  year = annees)

Nb_flower_IPM$estimated <- as.vector(t(Nf_est_IPM))
Nb_flower_IPM$observed[Nb_flower_IPM$Pop=="E2"] <- FloweringPlants$E2
```

```
## Warning: Unknown or uninitialised column: `observed`.
```

```
Nb_flower_IPM$observed[Nb_flower_IPM$Pop=="E1"] <- FloweringPlants$E1
Nb_flower_IPM$observed[Nb_flower_IPM$Pop=="Au"] <- FloweringPlants$Au
Nb_flower_IPM$observed[Nb_flower_IPM$Pop=="Po"] <- FloweringPlants$Po
Nb_flower_IPM$observed[Nb_flower_IPM$Pop=="Pe"] <- FloweringPlants$Pe
Nb_flower_IPM$observed[Nb_flower_IPM$Pop=="Cr"] <- FloweringPlants$Cr
Nb_flower_IPM <- Nb_flower_IPM %>%
  group_by(Pop) %>%
  mutate(lim = max(observed,estimated,na.rm=TRUE))

r2_data_IPM <- Nb_flower_IPM %>%
  group_by(Pop) %>%
  summarise(
    r2 = summary(lm(estimated ~ observed))$r.squared,
    biais = mean(estimated - observed, na.rm = TRUE),
    biais_relatif = 100 * biais / mean(observed, na.rm = TRUE))

labels_r2_IPM <- paste0(r2_data_IPM$Pop, " (R² = ", round(r2_data_IPM$r2, 2), ")")
```

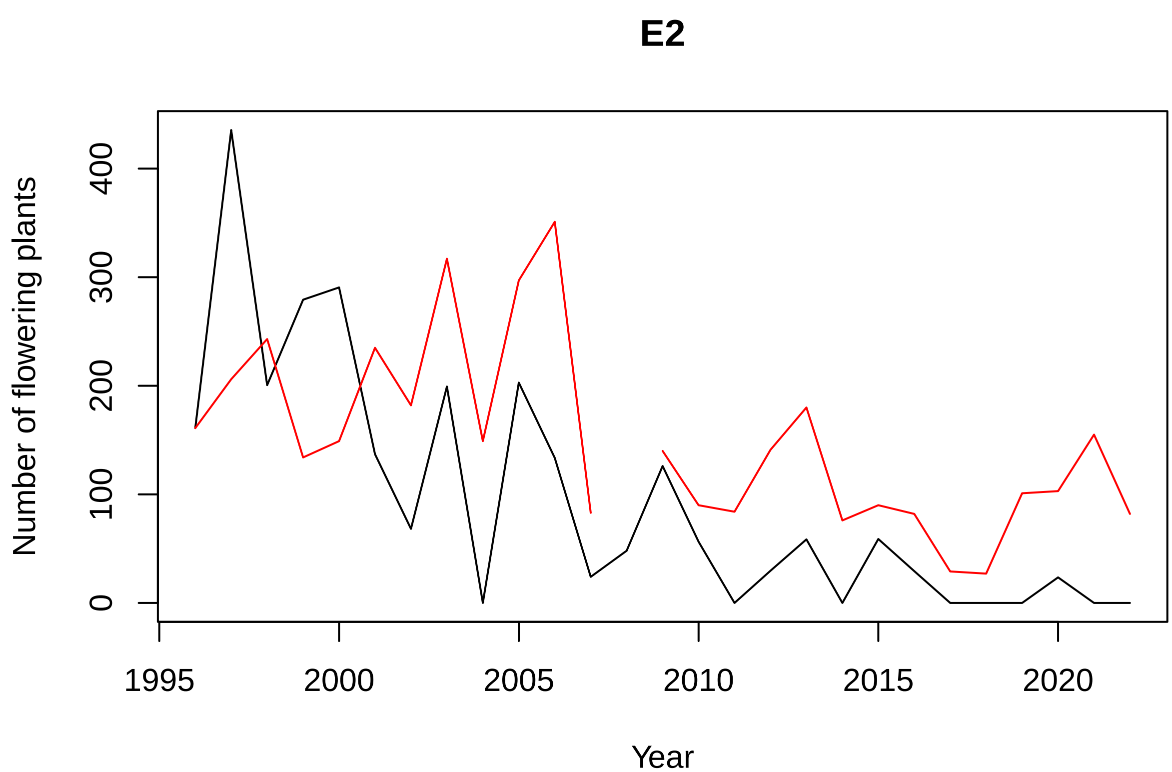
```r
ggplot(Nb_flower_IPM, aes(x = observed, y = estimated)) +
  geom_point(size = 2, alpha = 0.8, aes(color = as.factor(Pop))) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed") +
  labs(y = "Estimated number of flowering plants",
       x = "Observed number of flowering plants",
    color = "Population") +
  theme_minimal() +
  scale_color_viridis_d(
    option = "plasma",
    labels = labels_r2_IPM)+
  facet_wrap(~Pop,scales="free")+
  geom_blank(aes(x = lim, y = lim))
```

```
## Warning: Removed 6 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

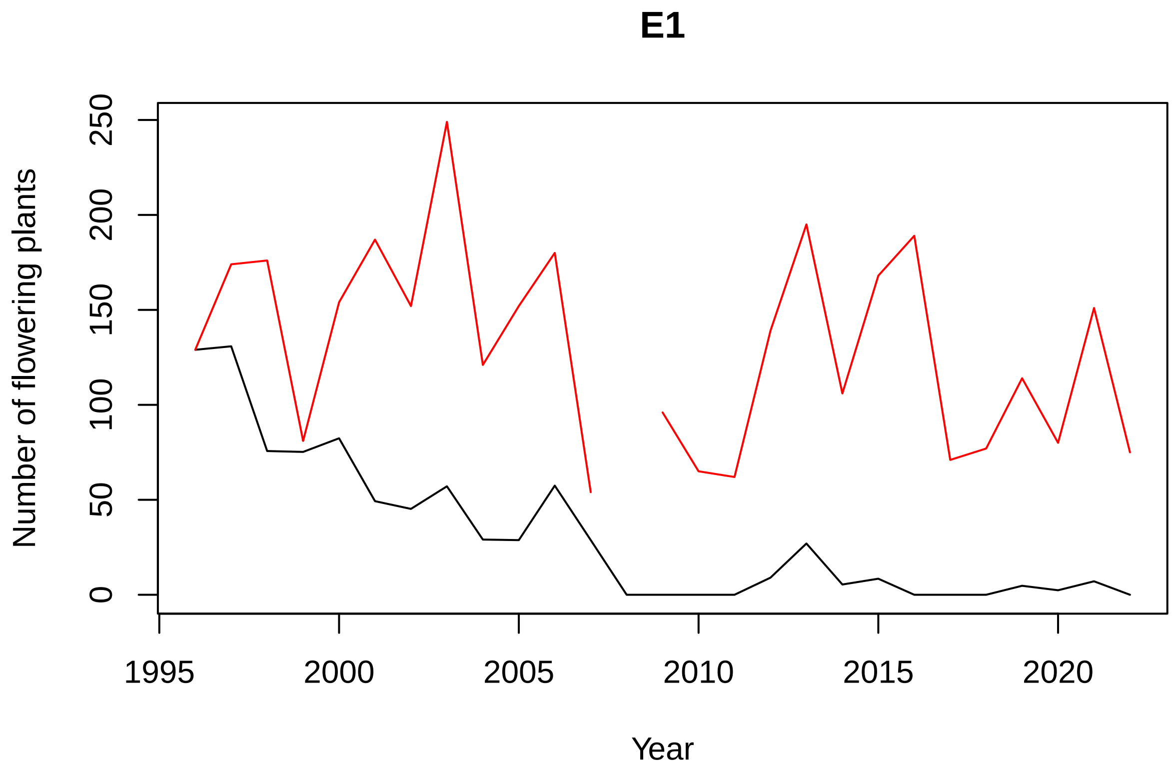



```r
load("../MPM/Aumatrices.Rdata");matricesAu <- matrices
load("../MPM/Crmatrices.Rdata");matricesCr <- matrices
load("../MPM/E1matrices.Rdata");matricesE1 <- matrices
load("../MPM/E2matrices.Rdata");matricesE2 <- matrices
load("../MPM/Pematrices.Rdata");matricesPe <- matrices
load("../MPM/Pomatrices.Rdata");matricesPo <- matrices

Nf_est_MPM <- matrix(NA, nrow=length(populations), ncol=length(annees))

for (p in 1:length(populations)) {
```

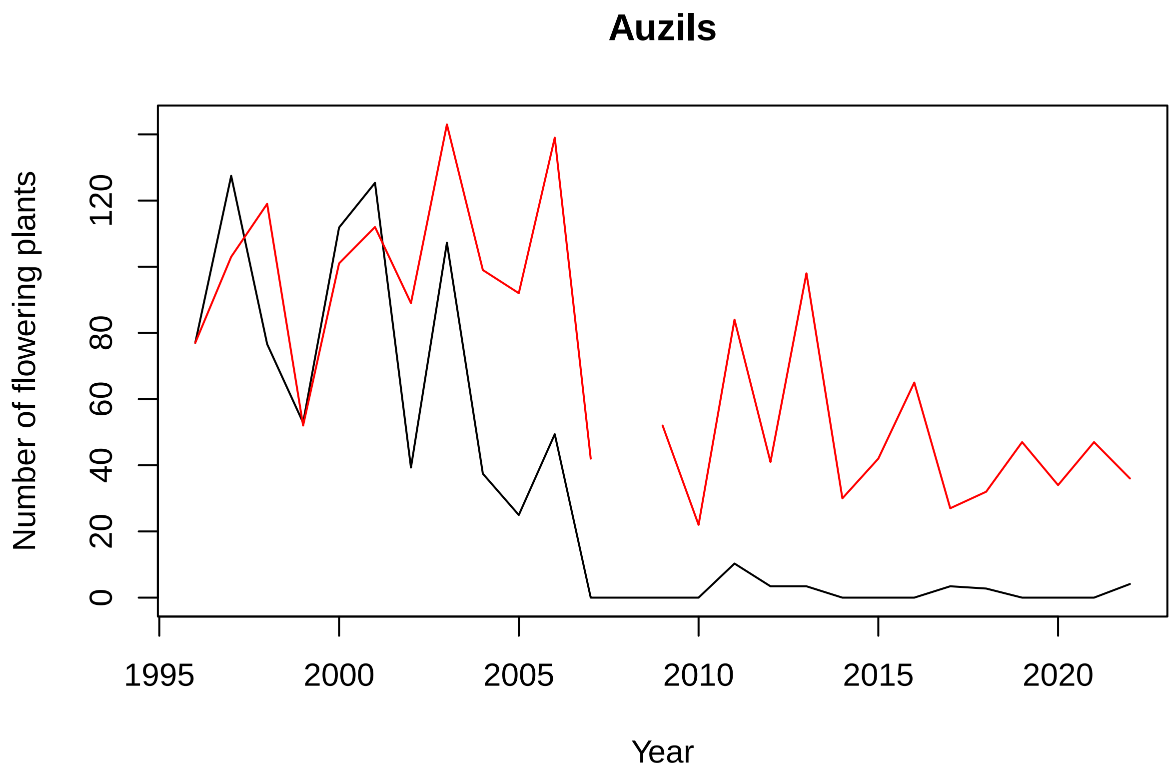
```
  Pop <- populations[p]
  if (Pop=="E2") {matrices <- matricesE2;Nf <- FloweringPlants$E2}
  if (Pop=="E1") {matrices <- matricesE1;Nf <- FloweringPlants$E1}
  if (Pop=="Au") {matrices <- matricesAu;Nf <- FloweringPlants$Au}
  if (Pop=="Po") {matrices <- matricesPo;Nf <- FloweringPlants$Po}
  if (Pop=="Pe") {matrices <- matricesPe;Nf <- FloweringPlants$Pe}
  if (Pop=="Cr") {matrices <- matricesCr;Nf <- FloweringPlants$Cr}

   ev <- eigen(mean(matrices)) #will be used to define distribution at t=0
   W <- abs(Re(ev$vectors[,1]))
   W <- W/sum(W)
   N <- round(W*Nf[1]/W[3])
   Nf_est_MPM[p,1] <- Nf[1]

   for (i in 2:length(annees)){
   M <- matrices[[i]]
   ev <- eigen(M)
   N <- M%*%N
   Nf_est_MPM[p,i] <- N[3]
  }
}


plot(y=Nf_est_MPM[1,], x=annees, type="l", ylim=c(0,max(c(Nf_est_MPM[1,],FloweringPlants$E2),na.rm=TRUE
```



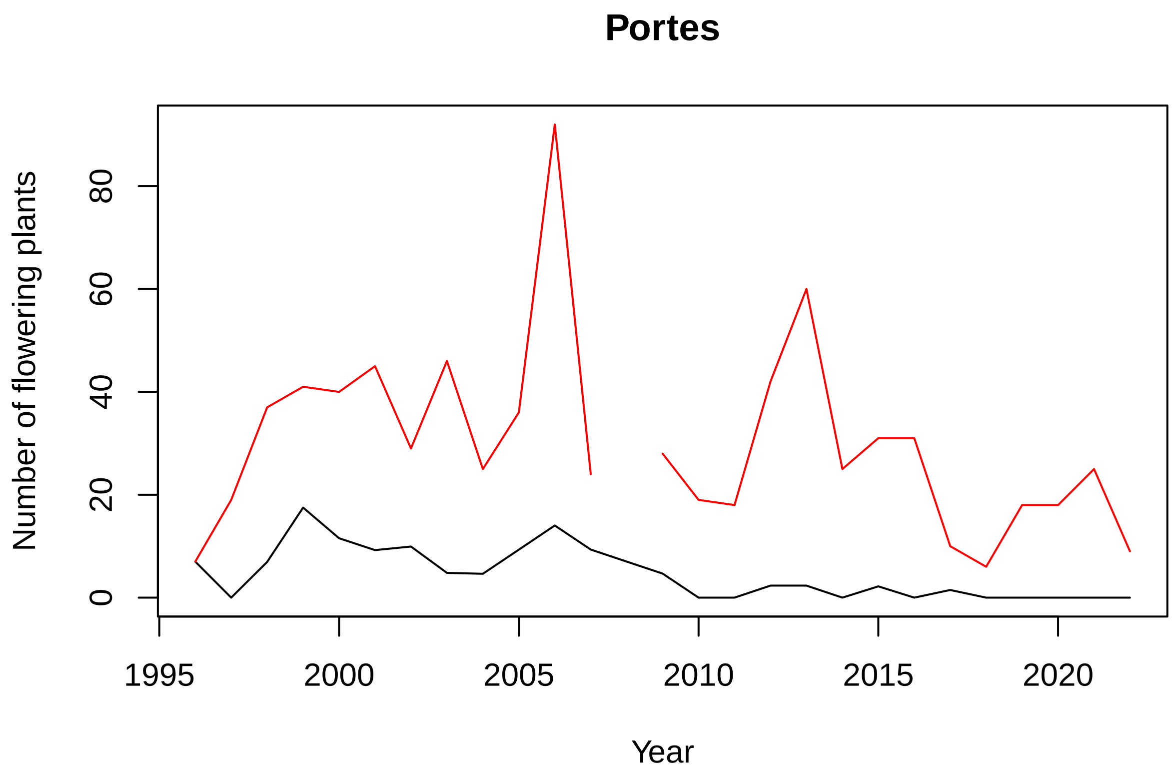
```r
plot(y=Nf_est_MPM[2,], x=annees,type="l",ylim=c(0,max(c(Nf_est_MPM[2,],FloweringPlants$E1),na.rm=TRUE))
```



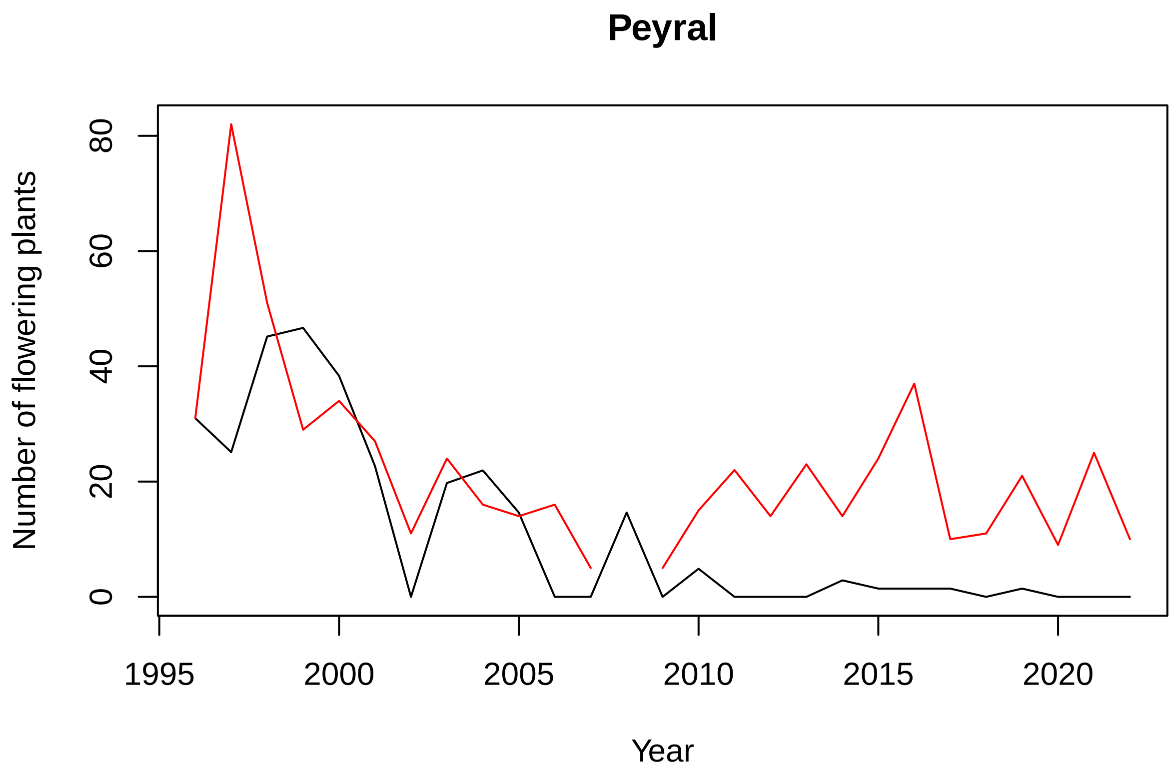

```r
plot(y=Nf_est_MPM[3,], x=annees,type="l",ylim=c(0,max(c(Nf_est_MPM[3,],FloweringPlants$Au),na.rm=TRUE))
```



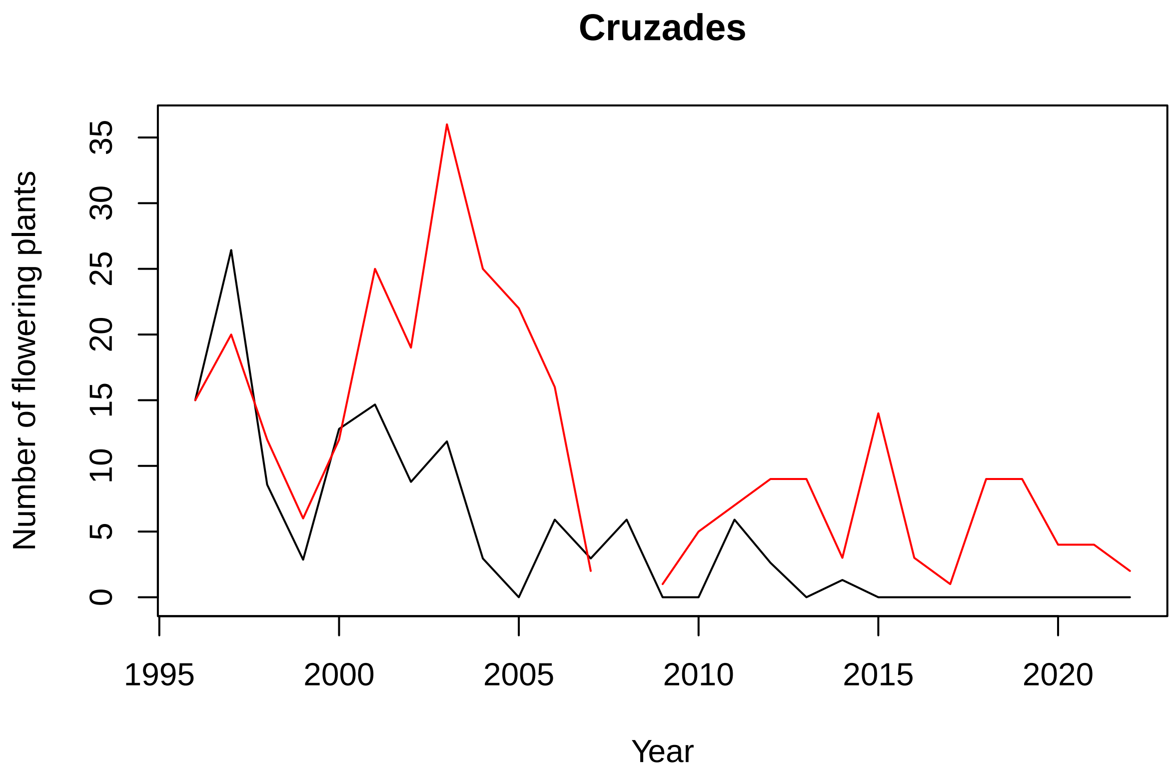

```
plot(y=Nf_est_MPM[4,], x=annees,type="l",ylim=c(0,max(c(Nf_est_MPM[4,],FloweringPlants$Po),na.rm=TRUE))
```




```
plot(y=Nf_est_MPM[5,], x=annees,type="l",ylim=c(0,max(c(Nf_est_MPM[5,],FloweringPlants$Pe),na.rm=TRUE))
```



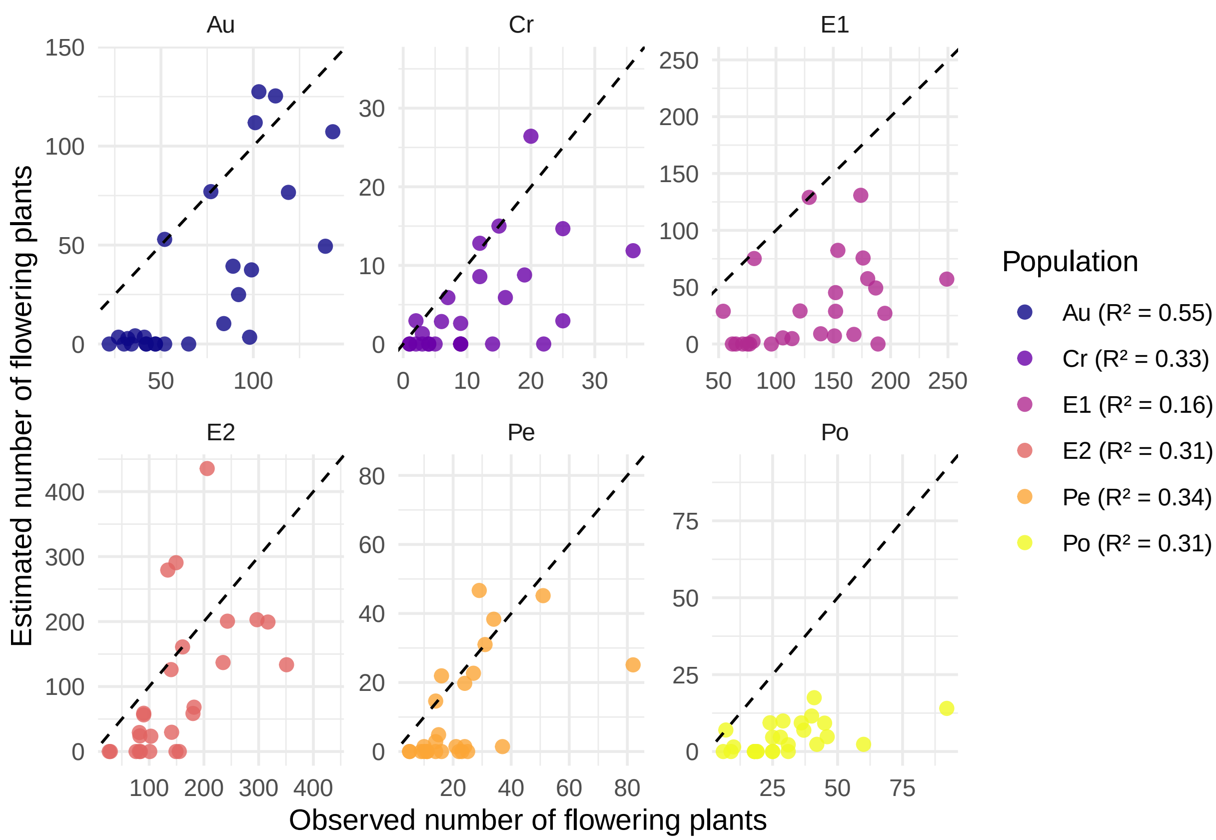

```
plot(y=Nf_est_MPM[6,], x=annees,type="l",ylim=c(0,max(c(Nf_est_MPM[6,],FloweringPlants$Cr),na.rm=TRUE))
```



Stat

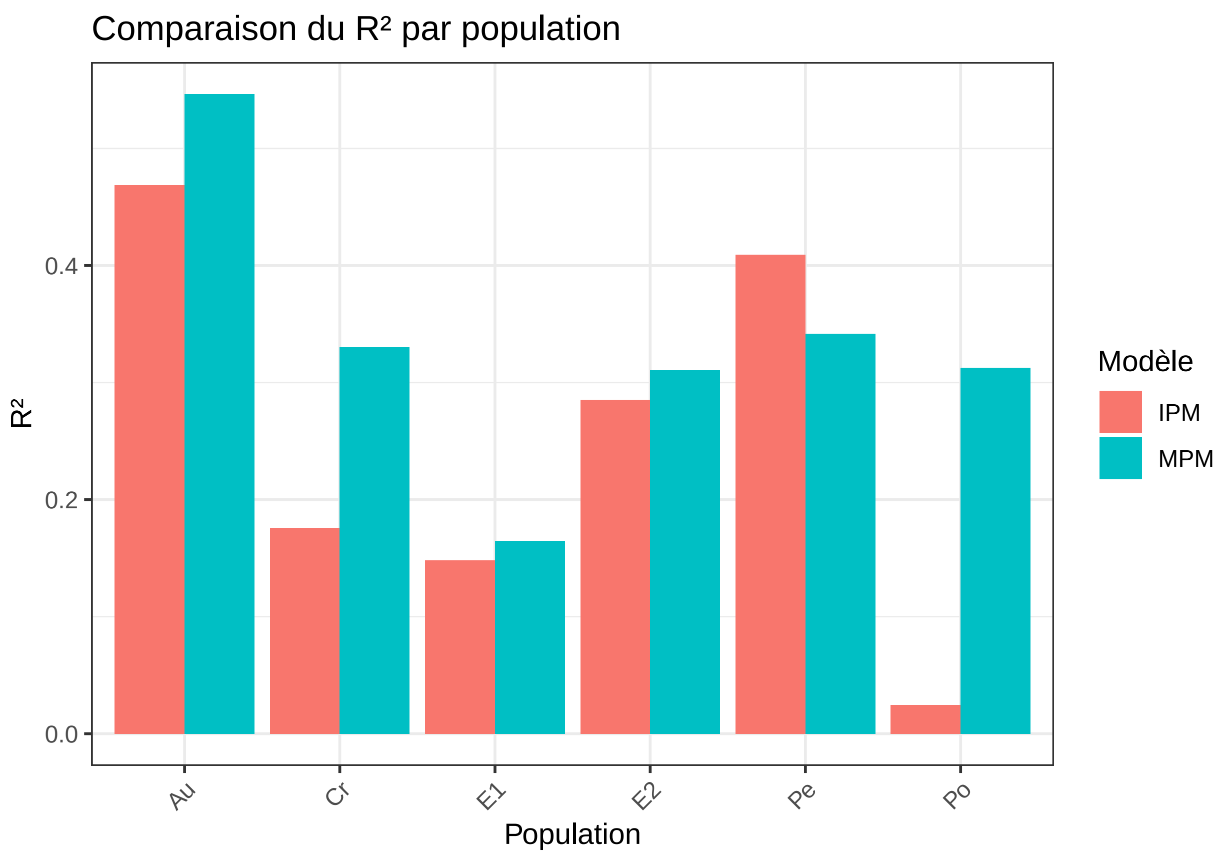
```
Nb_flower_MPM <- expand_grid(
  Pop = populations,
  year = annees)

Nb_flower_MPM$estimated <- as.vector(t(Nf_est_MPM))
Nb_flower_MPM$observed[Nb_flower_MPM$Pop=="E2"] <- FloweringPlants$E2
Nb_flower_MPM$observed[Nb_flower_MPM$Pop=="E1"] <- FloweringPlants$E1
Nb_flower_MPM$observed[Nb_flower_MPM$Pop=="Au"] <- FloweringPlants$Au
Nb_flower_MPM$observed[Nb_flower_MPM$Pop=="Po"] <- FloweringPlants$Po
Nb_flower_MPM$observed[Nb_flower_MPM$Pop=="Pe"] <- FloweringPlants$Pe
Nb_flower_MPM$observed[Nb_flower_MPM$Pop=="Cr"] <- FloweringPlants$Cr
Nb_flower_MPM <- Nb_flower_MPM %>%
  group_by(Pop) %>%
  mutate(lim = max(observed,estimated,na.rm=TRUE))

r2_data_MPM <- Nb_flower_MPM %>%
  group_by(Pop) %>%
  summarise(
    r2 = summary(lm(estimated ~ observed))$r.squared,
    biais = mean(estimated - observed, na.rm = TRUE),
    biais_relatif = 100 * biais / mean(observed, na.rm = TRUE))

labels_r2_MPM <- paste0(r2_data_MPM$Pop, " (R² = ", round(r2_data_MPM$r2, 2), ")")

ggplot(Nb_flower_MPM, aes(x = observed, y = estimated)) +
```

```
  geom_point(size = 2, alpha = 0.8, aes(color = as.factor(Pop))) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed") +
  labs(y = "Estimated number of flowering plants",
       x = "Observed number of flowering plants",
    color = "Population") +
  theme_minimal() +
  scale_color_viridis_d(
    option = "plasma",
    labels = labels_r2_MPM)+
  facet_wrap(~Pop,scales="free")+
  geom_blank(aes(x = lim, y = lim))
```

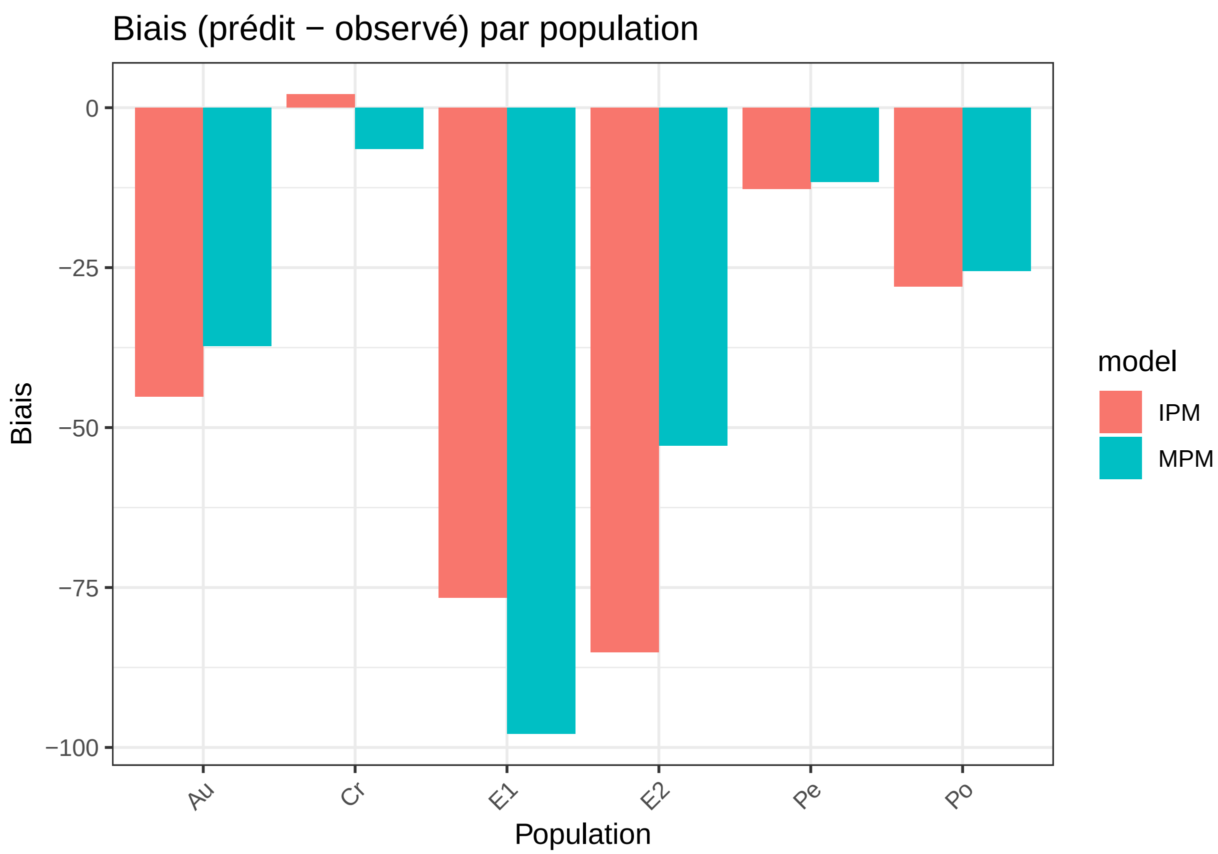


Comparaison IPM versus MPM

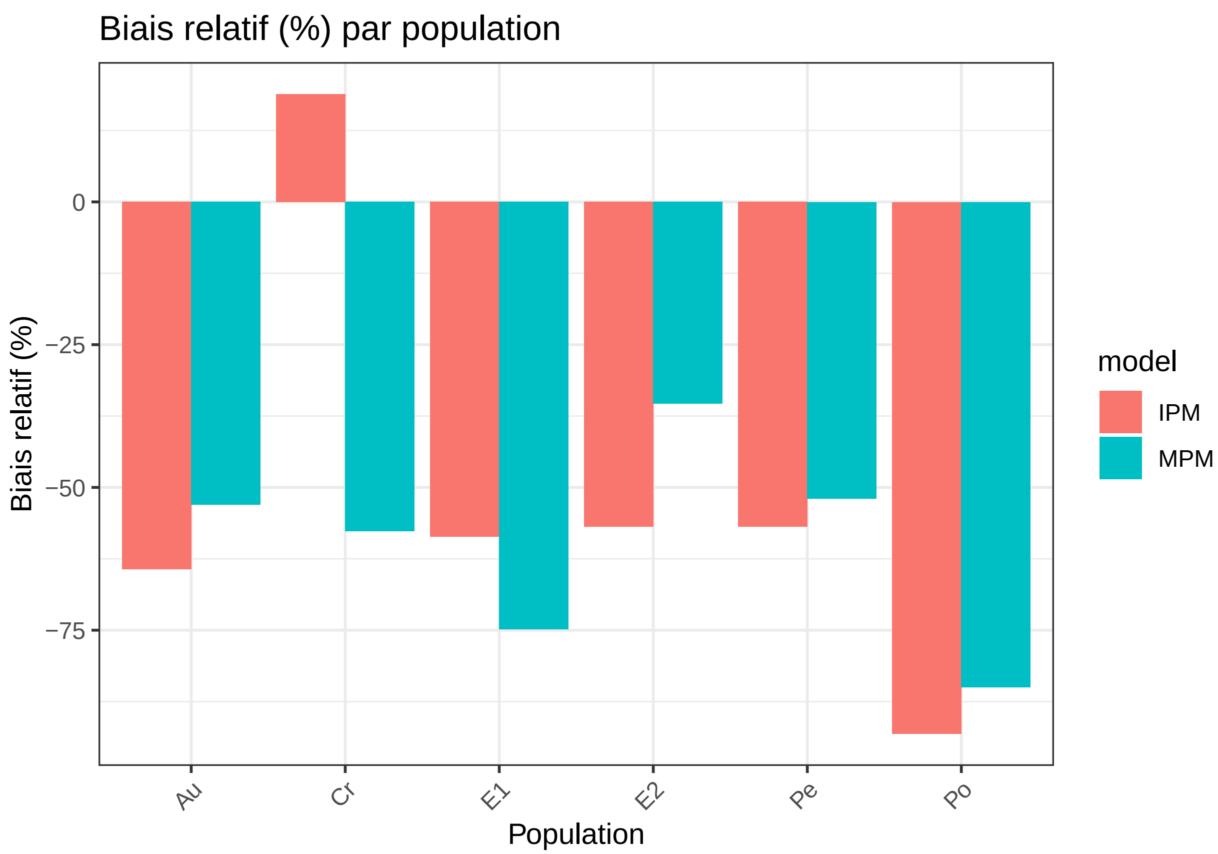
```
r2_data_IPM <- r2_data_IPM %>% mutate(model = "IPM")
r2_data_MPM <- r2_data_MPM %>% mutate(model = "MPM")
r2_data_joined <- bind_rows(r2_data_IPM, r2_data_MPM)


ggplot(r2_data_joined, aes(x = Pop, y = r2, fill = model)) +
  geom_col(position = "dodge") +
  theme_bw() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(title = "Comparaison du R² par population",
       x = "Population", y = "R²", fill = "Modèle")
```




```r
ggplot(r2_data_joined, aes(x = Pop, y = biais, fill = model)) +
  geom_col(position = "dodge") +
  theme_bw() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(title = "Biais (prédit - observé) par population",
       x = "Population", y = "Biais")
```




```r
ggplot(r2_data_joined, aes(x = Pop, y = biais_relatif, fill = model)) +
  geom_col(position = "dodge") +
  theme_bw() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(title = "Biais relatif (%) par population",
       x = "Population", y = "Biais relatif (%)")
```


Biais relatif (%) par population
Biais relatif (%)
0
−25
−50
−75
Au
Cr
E1
E2
Pe
Po
Population
model
IPM
MPM